

GMRCK25S Rear Helper Spring Kit Installation Manual

Model: GMRCK25S (Rear Suspension Package)

Please read these instructions carefully before starting the installation. Ensure all standard safety precautions are met while working underneath the vehicle.

1. Parts List

Before beginning, verify that all components listed in the table below are present in your kit.

Ref #	Description / Specifications	Qty
1	Hex Cap Screw (HCS) 3/8" x 1-1/4" UNF Grade 5 Zinc	4
2	Weldment Bracket (Left-Hand & Right-Hand)	1L - 1R
3	Flat Washer USS 3/8" x 1" Grade 2 Zinc	4
4	Nylock Locknut 3/8" UNF Grade 5 Zinc	4
5	Rubber Spacer (4" O.D. x 1" Thick with AH Bushing)	2
6	Hollow Rubber Spring	2
7	Hex Cap Screw (HCS) 3/8" x 2-3/4" UNC Grade 5 Zinc	2
8	Hex Cap Screw (HCS) 3/8" x 1-3/4" UNC Grade 5 Zinc	2

2. Installation Instructions

- 1. Raise Vehicle:** Raise the vehicle using an appropriate hoist or jack, allowing the axle to hang freely. Support the chassis securely before proceeding.
- 2. Remove Bump Stop:** Remove the existing OEM vehicle bump stop and its associated bracket.
- 3. Assemble Helper Spring Unit:** Assemble the hollow rubber spring and the weldment bracket. Refer to any provided diagrams for correct orientation.
- 4. Mount Assembly:** Install the complete helper spring assembly onto the vehicle frame utilizing the existing mounting holes and the provided fasteners.
- 5. Lower Vehicle:** Carefully lower the vehicle back to the ground.

Installation Notes:

- The rubber spacer (Ref 5) and the longer hex cap screw (Ref 7) may be utilized to adjust the installed height of the helper spring, if required, to achieve the desired ride height.
- Verify that vehicle brake lines or cables do not interfere with the helper spring's position or movement.
- Ensure the weldment bracket (Ref 2) is oriented with the larger disc side facing towards the vehicle's tire.

3. Clearance Instructions & Fitting Guidance

- **Clearance Gap (Light-Duty):** For light-duty trucks and vans, an ideal clearance of 1/2" to 1-1/2" should be maintained between the helper spring and the vehicle's contact point when the vehicle is unloaded.
- **Load Effects:** The presence of a partial load, cargo, or aftermarket vehicle accessories will reduce or eliminate this initial clearance gap.

- Heavy-Duty Applications: For medium and heavy-duty trucks, trailers, and motorhomes, the helper spring is designed to be in constant contact with the vehicle's frame, exhibiting no gap.

Important Safety Warnings:

- Do not exceed the vehicle manufacturer's recommended axle weight rating.
- Inspect for potential interference between the hollow rubber spring and other vehicle components. Under load, the rubber spring will expand in diameter.

Note: Illustrations are for reference only. If no specific reference image is provided, please install according to the standard mechanical guidelines.